Healthcare Funding & Outcomes Analysis

Visualizing the Landscape of Healthcare Research Funding

Submitted by: Anjali Sinha | ID: CX-138-MOR-JUNE

Project Foundations

Transforming complex multi-variable public health financial datasets into interactive visual dashboards using Tableau.

Executive Summary & Scope

Context & Challenge

Public health management relies on tracking how financial capital flows across medical research domains. Analyzing decades of funding records (2008–2024) alongside mortality and disease prevalence requires robust data architecture.

Methodological Approach

Integrating historical data, federal legislative injections (ARRA), and predictive modeling to align resource allocation with national health burdens, equipping policymakers with empirical clarity.

Strategic Analytical Objectives

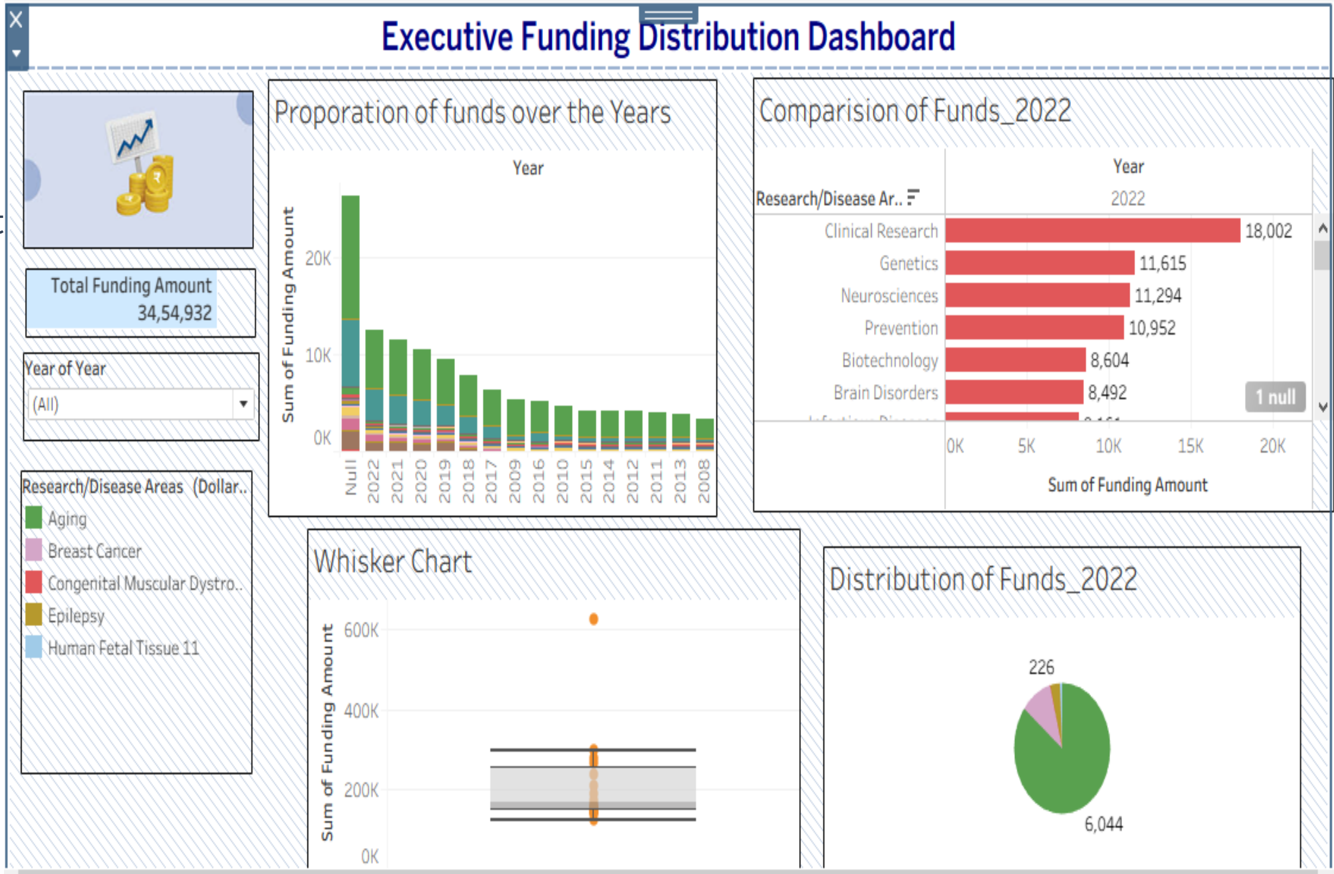
- **Trend Analysis:** Track historical shifts in funding (2008–2024) across distinct disease categories to evaluate consistent institutional support.
- **Comparative Evaluation:** Assess annual funding distributions and structural priorities between competing medical research domains.
- **Outcome Correlation:** Analyze statistical relationships between research expenditures and real-world health indicators like 2019 US mortality.
- **Economic Impact:** Evaluate how major stimulus packages (e.g., ARRA) accelerated development across target medical sectors.

Dashboard Insights & Metrics

Evaluating funding distributions, high-priority allocations, and distribution anomalies.

Funding Volume & Yearly Proportions

- **Cumulative Capital:** Total recorded funding stands at **34,54,932** (Indian numbering system format).
- **Historical Decay:** Defined yearly funding peaks strongly in recent cycles (2022 clearing 10K) and progressively decreases toward 2008.
- **Data Integrity Note:** A prominent "Null" category spike indicates unclassified records lacking proper year-tags.



2022 Research Domain Priorities

Clinical Research commands the clear lead in 2022 resource allocation, closely backed by foundational biosciences like Genetics and Neurosciences.

Distribution Spread & Outlier Analysis

The Whisker Distribution

The vast majority of project funding data points cluster tightly between the 100K and 300K thresholds within the standard quartile boundaries.

Extreme Funding Outlier

A highly prominent statistical anomaly sits past **600K** on the Y-axis, representing a singular mega-grant or institutional initiative far outside typical variance.

Disease-Specific & Intensity Analytics

Isolating high-burden medical conditions and examining multi-year financial trajectories.

Longitudinal Tracking: Liver Cancer

- **Growth Trajectory:** Funding starts at 89 (2008), experiences a valley between 2011–2014 (low of 71), and enters a sustained post-2015 upward climb to peak at 178 in 2022.
- **Policy Correlation:** Post-2015 increases reflect targeted federal attention toward hepatic disease burdens and diagnostic improvements.

Funding Intensity Across Disease Sectors

Research / Disease Area	Funding Intensity ($ Millions Rounded)
Aging	**66,402** (Dominant Tier)
Acquired Cognitive Impairment	24,865
Alzheimer's Disease (incl. AD/ADRD)	24,499
Alcoholism, Alcohol Use and Health	8,891

Strategic Insights & Recommendations

Financial Resilience

Sustained capital commitment in sectors like Aging and Clinical Research demonstrates robust institutional resilience against macroeconomic volatility.

Optimizing Future Allocations

Bridging geographic and categorical data gaps ensures future capital models directly address true national mortality burdens and regional care disparities.

Questions?

Thank you for your attention. Open for discussion.

Healthcare Funding & Outcomes Analysis | Anjali Sinha